THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS FULL SYLLABUS PAPER SET 09

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. $\sqrt[3]{729}$ is: 1

(a) 10 **(b)** 9 **(c)** 8 **(d)** 11

2. Which of the following is a perfect square? 1

(a) 500 **(b)** 400 **(c)** 450 **(d)** 410

3. The number 4500 in standard (scientific) form is: 1

(a) 4.5×10^4 **(b)** 0.45×10^3 **(c)** 45×10^3 **(d)** 4.5×10^3

4. The Maya number system was based on: 1

(a) 10 **(b)** 60 **(c)** 12 **(d)** 20

5. The diagonal of a rectangle with sides 6 cm and 8 cm is: 1

(a) 10 cm **(b)** 14 cm **(c)** 9 cm **(d)** 11 cm

6. In rhombus ABCD, diagonal AC bisects angle: 1

(a) A and C **(b)** D only **(c)** B only **(d)** none

7. A number is divisible by 4 if the number formed by its last two digits is divisible by: 1

(a) 5 **(b)** 4 **(c)** 2 **(d)** 8

8. $(a-b)^2$ equals: 1

(a) $a^2+2ab+b^2$ **(b)** $a^2-2ab+b^2$ **(c)** a^2-b^2 **(d)** a^2-ab+b^2

9. If 7 : 8 = 21 : x, then x equals: 1

(a) 24 **(b)** 27 **(c)** 23 **(d)** 25

10. The ratio of 15 minutes to 2 hours is: 1

(a) 1 : 4 **(b)** 1 : 6 **(c)** 15 : 2 **(d)** 1 : 8

11. GST is a tax charged on: 1

(a) land only **(b)** property transfer only

(c) income only **(d)** goods and services

12. $\frac{9}{25}$ expressed as a percentage is: 1

(a) 31% **(b)** 36% **(c)** 41% **(d)** 72%

13. If the diagonal of a square is 10 cm, its side is: 1

(a) 20 cm **(b)** 5 cm **(c)** 10 cm **(d)** $5\sqrt{2}$ cm

14. The legs of a right triangle are 12 cm and 16 cm. Its hypotenuse is: 1

(a) 20 cm **(b)** 19 cm **(c)** 28 cm **(d)** 21 cm

15. ₹ 1100 is shared among A, B and C in the ratio 2:3:6. B gets: 1

(a) ₹ 366 **(b)** ₹ 300 **(c)** ₹ 600 **(d)** ₹ 200

16. The Koch snowflake is built by repeatedly adding triangles to: 1

(a) the base only **(b)** the interior

(c) the vertices **(d)** the middle third of each side

17. The mean of the data 8, 8, 10, 12, 14, 18 is: 1

(a) 12 **(b)** 13.6667 **(c)** 11.6667 **(d)** 9.66667

18. In a number pyramid, each block is the sum of the two below it. The bottom row is 8, 1, 9. The top block is: 1

(a) 20 **(b)** 21 **(c)** 18 **(d)** 19

19. The area of a rhombus with diagonals 24 cm and 10 cm is: 1

(a) 34 cm² **(b)** 120 cm² **(c)** 240 cm² **(d)** 60 cm²

20. A rectangle has area 84 cm² and length 7 cm. Its breadth is: 1

(a) 11 cm **(b)** 77 cm **(c)** 14 cm **(d)** 12 cm

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. State the law for multiplying powers with the same base. Give an example. 2

22. Convert 19 to base 2. 2

23. Check whether 7392 is divisible by 8. 2

24. If 8 workers finish a job in 15 days, how many days will 12 workers take? 2

25. What is the converse of the theorem? 2

26. Divide 360 in the ratio 1 : 2 : 3. 2

27. Find the median of 7, 3, 9, 5, 12, 15, 1. 2

28. Find the area of a trapezium with parallel sides 9 cm, 13 cm and height 7 cm. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Find the smallest square number divisible by 10, 15 and 20. 3

30. Explain with a figure idea why a trapezium is not a parallelogram. 3

31. Expand and simplify $(x + 5)(x + 7)$. Verify for $x = 1$. 3

32. A man saves ₹ 4500 per month, which is 15% of his salary. Find his salary and annual savings. 3

33. A cube of edge 3 cm is painted and cut into 27 unit cubes. How many cubes have exactly two faces painted? 3

34. Prove that the sum of any three consecutive even numbers is a multiple of 6. 3

35. A ladder of length 10 m rests against a wall with its foot 6 m from the wall. How high up the wall does it reach? 3

OR

A rectangular park measures 20 m by 21 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

36. Find the area of a trapezium with parallel sides 14 cm and 22 cm and distance between them 7 cm. Also find the area of an equal-height rectangle with width 22 cm. 3

OR

A rectangular plot is 14 m by 9 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) The diameter of a hydrogen atom is about 10^{-10} m and of a pollen grain about 10^{-5} m. How many times larger is the pollen grain? 5
(ii) Express a red blood cell of 7×10^{-6} m in mm.

OR

(i) Why do scientists prefer scientific notation? 5
(ii) Write the mass of an electron, 0.00000000000000000000000000091 kg, in scientific notation.

38. (i) Express the ratio 3 : 5 as percentages. 5
(ii) A mixture is 35% acid; how much acid in 80 L?

OR

(i) Show that two successive increases of 10% equal one increase of 21%. 5
(ii) What about 10% increase followed by 10% decrease?

39. **Case study:** A cricketer's runs in 8 matches: 12, 45, 0, 38, 62, 18, 45, 90. 5
(i) Find mean and median.
(ii) Find range.
(iii) Comment.

OR

(i) If each of 9 observations is multiplied by 2, how do the mean and median change? 5
(ii) Illustrate with 3, 5, 7.

40. (i) Find the area of an L-shaped region made of rectangles 10×4 and 6×3. 5
(ii) Suggest another method.

OR

(i) Show that the area of a rhombus is $\frac{1}{2}d_1d_2$. 5
(ii) Find the diagonal if the area is 120 cm² and the other diagonal is 15 cm.

END OF QUESTION PAPER — Total: 80 Marks

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